Twisted-Convolution Identities in Dimensions Seven and Eight: Shintani Height Rigidity and CM Descent

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Abstract

We prove the formal Twisted Convolution Conjecture of Appleby–Flammia–Kopp unconditionally for every admissible tuple in dimensions seven and eight. Dimension seven has two form conductors, and we prove the discriminant-8 and discriminant-32 strata independently. For discriminant 32, conductor lowering places every characteristic in one of six maximal-order ray groups. For discriminant 8, the direct six-factor Shintani–Faddeev cocycle has alternating periods $2+\sqrt{2}$ and $(2+\sqrt{2})/2$. Shintani's proved index-two algebraicity theorem gives a common safe exponent 16128; rigorous Arb enclosures, Voutier's height gap, exact phase and Artin labels, and degree-48 compositum certificates identify both complete signed overlap packets and make all 441 rank-two minors vanish for both formal shifts in each stratum. Dimension eight has two admissible form conductors. For conductor three, genuine linear reinduction to quartic ray characters over $\mathbb{Q}(\sqrt{-6})$ and Stark's proved imaginary-quadratic rank-one theorem determine the two formerly missing orientations. For conductor one, the relevant ray characters are quadratic; analytic class-number formulas determine their units, and an exact phase calculation in $\mathbb{Q}(\sqrt{5})$ orients all 63 nonzero cocycle values. Exact quotient-ring arithmetic makes all 784 minors vanish for both shifts in each dimension-eight stratum. No unproved Stark conjecture or numerical recognition is used as a logical input.

1 Formal TCC and conventions

We use the conventions of Appleby–Flammia–Kopp (AFK) [1]. An admissible tuple $t=(d,r,Q)$ contains a primitive indefinite integral binary quadratic form Q, a positive real fixed point ρ_t, a level-d stabilizer A_t, and the matrix $L_{z,t}$. For $\boldsymbol{p}\in\mathbb{Z}^2$, let $\mathcal{I}_{\boldsymbol{p}}$ be a complete set of representatives modulo d containing $\mathbf{0}$ and $\boldsymbol{p}$. A residue $\lambda\in\mathbb{Z}/d\mathbb{Z}$ is a *shift* when its AFK coprimality condition holds and

$$\sum_{\boldsymbol{q}\in\mathcal{I}_{\boldsymbol{p}}}\omega_d^{r\langle\boldsymbol{p},(\lambda I+L_{z,t})\boldsymbol{q}\rangle}\operatorname{shin}_{A_t}^{\boldsymbol{q}/d}(\rho_t)\operatorname{shin}_{A_t^{-1}}^{(\boldsymbol{q}-\boldsymbol{p})/d}(\rho_t)=d^2\delta_{\boldsymbol{p},\mathbf{0}}^{(d)}\tag{TCC}$$

for every $\boldsymbol{p}$. Write $\mathcal{Z}_t$ for the shift set. The formal TCC asserts $0,1\in\mathcal{Z}_t$ and equality of shift sets for admissible forms of the same discriminant.

Our reciprocal double sine is

$$S_2(z\mid\omega_1,\omega_2):=\frac{\Gamma_2(z\mid\omega_1,\omega_2)}{\Gamma_2(\omega_1+\omega_2-z\mid\omega_1,\omega_2)}.\tag{1}$$

It is the reciprocal of Kopp's $\mathrm{Sin}_{2,\mathrm{Kopp}}$ convention [2, Definition 4.22]. In particular,

$$S_2(z+1\mid\beta,1)=2\sin(\pi z/\beta)S_2(z\mid\beta,1).\tag{2}$$

The place labels ∞_1,∞_2 always refer to the positive and negative square-root embeddings printed in the certificates. PARI orders the negative root first in the quadratic fields used below; all raw PARI conductor vectors are translated to the displayed place labels before use.

Proposition 1 (Finite reconstruction bridge)**.** *Suppose the convention-matched AFK values $\nu_{\boldsymbol{p}}$ are inserted into the normalized Weyl reconstruction for a permitted twist. Then the reconstructed matrix Π satisfies $\operatorname{Tr}\Pi = 1$, and equation* (TCC) *is equivalent to $\Pi^2 = \Pi$. If all rank-two minors vanish and $\Pi \neq 0$, then Π is a rank-one idempotent and the corresponding residue is a shift.*

Proof. Expand the Weyl coefficient of $\Pi^2 - \Pi$, use the Weyl multiplication law, and reindex by the displacement difference. The AFK endpoint correction changes the raw diagonal value $\sqrt{d+1}$ to the normalized zero overlap one. The resulting coefficient is exactly the left side of (TCC) minus its $d^2\delta$ term. Fourier inversion is injective. Finally, vanishing of the 2×2 minors gives rank at most one, while trace one gives rank exactly one and idempotency. □

Theorem 2. *For every admissible dimension-seven or dimension-eight tuple t,*

$$0, 1 \in \mathcal{Z}_t.$$

If t, t' have forms of the same discriminant, then $\mathcal{Z}_t = \mathcal{Z}_{t'}$.

2 Dimension seven

2.1 Admissible strata and scope

The admissibility recurrence forces

$$d = 7 \Longrightarrow r = 1, \quad n = 8, \quad K_7 = \mathbb{Q}(\sqrt{2}), \quad j = m = 1, \quad f_1 = 2.$$

The form conductor may therefore be $f = 1$ or $f = 2$. Both strata exist; convenient representatives are

$$Q_{7,1} = \langle 1, -4, 2 \rangle, \quad \operatorname{disc} Q_{7,1} = 8, \qquad Q_{7,2} = \langle 1, -6, 1 \rangle, \quad \operatorname{disc} Q_{7,2} = 32.$$

The two discriminants are distinct covariance strata, so they must be proved independently. We first prove the missing maximal-order tuple $Q_{7,1}$ and then recall the conductor-lowering proof for $Q_{7,2}$. Since both orders have wide class number one, covariance then covers every admissible form in each stratum.

2.2 The maximal-order discriminant-eight packet

Put

$$\alpha = 2 + \sqrt{2}, \qquad L_{7,1} = \begin{pmatrix} 7 & -4 \\ 2 & -1 \end{pmatrix}, \qquad A_{7,1} = L_{7,1}^3 = \begin{pmatrix} 239 & -140 \\ 70 & -41 \end{pmatrix}.$$

The first number is the positive fixed point of $Q_{7,1}$. Direct matrix multiplication gives $A_{7,1} \equiv I \pmod 7$, and the exact Dedekind-sum calculation gives $\Psi(A_{7,1}) = 0$. Thus the AFK phase is

$$\Phi^{(1)}_{\boldsymbol{p}} = -\tau_7^{-2Q_{7,1}(\boldsymbol{p})}, \qquad \tau_7 = -e^{\pi i/7}. \tag{3}$$

All of these assertions, together with all 48 inverse-ray labels, are replayed by the maximal-tuple audit script:

`scripts/dimension_seven_maximal_tuple_audit.gp`.

The canonical continued-fraction word for $A_{7,1}$ is

$$[4, 2, 4, 2, 4, 2, 0]. \tag{4}$$

Consequently the direct Shintani–Faddeev formula has six reciprocal double-sine factors, with periods alternating between α and $\alpha/2$. This is not the three-factor discriminant-32 formula with its fixed point relabeled.

The one-place and two-place ray groups modulo seven are respectively C_6 and $C_6 \times C_2$. The conjugation commutator and the kernel obtained by forgetting the second real place are distinct order-two subgroups. Hence the one-place Stark field has index two over its maximal absolutely abelian intersection, exactly the hypothesis of Shintani's theorem. It is a subfield of the ray-14 field used below. The full divisor and real-distribution audit for that field therefore supplies the same safe exponent

$$m = 16128. \tag{5}$$

The six nonquadratic signed values are the relevant real roots of

$$\begin{aligned} G(X) = X^{12} &+ 4X^{11} - 2X^{10} - 22X^9 - 18X^8 + 16X^7 \\ &+ 41X^6 + 16X^5 - 18X^4 - 22X^3 - 2X^2 + 4X + 1, \end{aligned} \tag{6}$$

while the two remaining nontrivial magnitudes are the negative real roots of

$$R(X) = X^4 + 2X^3 + X^2 + 2X + 1. \tag{7}$$

These are exact signed-square factorizations of the certified Stark-unit polynomials: no integer-relation output is used as a proof input. Both fields embed in the degree-24 ray-14 overlap field. Rational Sturm intervals isolate every displayed root.

The six-factor Arb evaluator `scripts/certify_dimension_seven_maximal_cocycle.py` gives

$$\epsilon_{\log} \le 2.30 \cdot 10^{-11}, \qquad 16128\epsilon_{\log} \le 3.71 \cdot 10^{-7}. \tag{8}$$

The complete Arb balls are archived in

`certificates/dimension-seven-`
`maximal-cocycle-intervals.txt`.

Voutier's lower bound is greater than $5.22 \cdot 10^{-5}$ in every relevant degree, so height rigidity identifies the analytic magnitudes with the isolated algebraic roots. Independently, the maximal sign-audit script

`scripts/dimension_seven_maximal_sign_audit.py`

tracks every finite q-Pochhammer phase and double-sine recurrence in $\mathbb{Q}(\sqrt{2})$; the irrational phase coefficient cancels in all 48 cases and fixes the complete sign table.

Finally, the exact maximal-order TCC script

`scripts/dimension_seven_maximal_exact_tcc.gp`

inserts the sixteen Zauner-orbit representatives using

$$(a, b) \longmapsto (3b, 2a - b) \pmod 7. \tag{9}$$

In the same degree-48 compositum with $\mathbb{Q}(\zeta_{56})$ used below, it proves for each formal shift

$$\operatorname{Tr}\Pi = 1, \qquad \Pi^2 = \Pi, \qquad 441 \text{ rank-two minors vanish.} \tag{10}$$

Proposition 1 proves both shifts for $Q_{7,1}$. The order of discriminant 8 has wide class number one, so covariance proves the entire discriminant-8 stratum.

2.3 Conductor lowering and ray classes

Put

$$K_7 = \mathbb{Q}(\sqrt{2}), \qquad \alpha = 2+\sqrt{2}, \qquad \beta_7 = 3+2\sqrt{2}.$$

Here β_7 is the positive fixed point of $Q_{7,2}$, whereas α is the reduced maximal-order point used only after conductor lowering. With

$$L_7 = \begin{pmatrix} 6 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_{7,2} = L_7^3 = \begin{pmatrix} 204 & -35 \\ 35 & -6 \end{pmatrix},$$

the exact Rademacher calculation gives

$$\Phi_{\boldsymbol{p}} = \zeta_{56}^{7-32Q_{7,2}(\boldsymbol{p})}.$$

The determinant-two map

$$B = \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}$$

sends α to β_7. For a characteristic $(a,b)/7$, its two preimages are

$$s_j = \frac{1}{14}\begin{pmatrix} a+b+7j \\ 2b \end{pmatrix}, \qquad j = 0,1. \tag{11}$$

Kopp's inverse ray map assigns to s_j the principal element

$$\gamma_{a,b,j} = 2b\sqrt{2} + 3b - a - 7j. \tag{12}$$

Removing the common ideal divisor of (14) and (γ) produces exactly six lowered moduli. Their one-place ray groups and Kopp exponents are

modulus HNF	ray group	n
$\left(\begin{smallmatrix} 14 & 0 \\ 0 & 14 \end{smallmatrix}\right)$	$C_6 \times C_2$	1
$\left(\begin{smallmatrix} 7 & 0 \\ 0 & 7 \end{smallmatrix}\right)$	C_6	1
$\left(\begin{smallmatrix} 14 & 6 \\ 0 & 2 \end{smallmatrix}\right)$	C_2	1
$\left(\begin{smallmatrix} 14 & 8 \\ 0 & 2 \end{smallmatrix}\right)$	C_2	1
$\left(\begin{smallmatrix} 7 & 3 \\ 0 & 1 \end{smallmatrix}\right)$	C_2	2
$\left(\begin{smallmatrix} 7 & 4 \\ 0 & 1 \end{smallmatrix}\right)$	1	2.

(13)

The common determinant-one stabilizer of the lowered factors at α is

$$A_{\mathrm{low}} = \begin{pmatrix} 239 & -140 \\ 70 & -41 \end{pmatrix}. \tag{14}$$

The exact inverse-ray, gcd, sign-class, and stabilizer computations are printed for all 96 lifted factors by `scripts/explore_dimension_seven_conductor_lowering.gp`.

2.4 Powered algebraicity and packet identification

The full one-place ray-14 field has relative group $C_6 \times C_2$. In its normal closure, the maximal absolutely abelian subfield has index two and the applicable imaginary-quadratic base is $\mathbb{Q}(\sqrt{-7})$. Auditing all 32 divisors of its conductor gives

$$m_{\mathfrak{d}} = \begin{cases} 12h_{\mathbb{Q}(\sqrt{-7})}n(S_{\mathfrak{d}}), & \mathfrak{d} = (1), \\ 12f_{\mathfrak{d}}n(S_{\mathfrak{d}}), & \mathfrak{d} \neq (1), \end{cases} \qquad m = \mathrm{lcm}_{\mathfrak{d}\mid\mathfrak{c}}\, m_{\mathfrak{d}} = 16128. \tag{15}$$

Write $\mathfrak{c}=\mathfrak{p}_2^3\bar{\mathfrak{p}}_2^3\mathfrak{p}_7$ and $\mathfrak{d}(e_1,e_2,e_7)=\mathfrak{p}_2^{e_1}\bar{\mathfrak{p}}_2^{e_2}\mathfrak{p}_7^{e_7}$. The complete divisor audit is below. Each triple records $(|H_\mathfrak{d}|, n(S_\mathfrak{d}), m_\mathfrak{d})$; here $|H_\mathfrak{d}|$ is the ray-group order, and w_0/w_1 records $w(\mathfrak{d})$ for $e_7=0/1$. The two triple columns have $e_7=0$ and $e_7=1$, respectively.

e_1	e_2	w_0/w_1	$e_7=0$	$e_7=1$
0	0	2/1	(1, 96, 1152)	(3, 16, 1344)
0	1	2/1	(1, 96, 2304)	(3, 16, 2688)
0	2	1/1	(1, 48, 2304)	(6, 8, 2688)
0	3	1/1	(2, 24, 2304)	(12, 4, 2688)
1	0	2/1	(1, 96, 2304)	(3, 16, 2688)
1	1	2/1	(1, 96, 2304)	(3, 16, 2688)
1	2	1/1	(1, 48, 2304)	(6, 8, 2688)
1	3	1/1	(2, 24, 2304)	(12, 4, 2688)
2	0	1/1	(1, 48, 2304)	(6, 8, 2688)
2	1	1/1	(1, 48, 2304)	(6, 8, 2688)
2	2	1/1	(2, 24, 1152)	(12, 4, 1344)
2	3	1/1	(4, 12, 1152)	(24, 2, 1344)
3	0	1/1	(2, 24, 2304)	(12, 4, 2688)
3	1	1/1	(2, 24, 2304)	(12, 4, 2688)
3	2	1/1	(4, 12, 1152)	(24, 2, 1344)
3	3	1/1	(8, 6, 576)	(48, 1, 672)

Thus the lcm of all thirty-two displayed exponents is 16128. The full HNF ideals and independently recomputed ray orders are in the archived transcript

`certificates/dimension-seven-shintani-divisors.txt`.

Here the unit-congruence factor $w(\mathfrak{d})$ is computed separately for every divisor; four divisors, including the trivial one, have $w=2$. Shintani's proved theorem [4] therefore makes the powered packet algebraic.

The sixteen squared-overlap representatives satisfy the reciprocal polynomial

$$\begin{aligned} P(X) = X^{16} &- (37+28\sqrt{2})X^{15} + (1212+854\sqrt{2})X^{14} \\ &- (20685+14630\sqrt{2})X^{13} + (210371+148750\sqrt{2})X^{12} \\ &- (1313872+929054\sqrt{2})X^{11} + (5031845+3558044\sqrt{2})X^{10} \\ &- (11531249+8153832\sqrt{2})X^{9} + (15288774+10810788\sqrt{2})X^{8} \\ &- (11531249+8153832\sqrt{2})X^{7} + (5031845+3558044\sqrt{2})X^{6} \\ &- (1313872+929054\sqrt{2})X^{5} + (210371+148750\sqrt{2})X^{4} \\ &- (20685+14630\sqrt{2})X^{3} + (1212+854\sqrt{2})X^{2} \\ &- (37+28\sqrt{2})X + 1. \end{aligned}$$

It factors over $\mathbb{Q}(\sqrt{2})$ with degrees $2, 2, 12$. Exact class-field comparisons identify those factors with the two quadratic lowered fields P_+, P_- and the one-place ray-14 field. The following rational intervals each contain exactly one real root by Sturm's theorem; the last column is the exact ray log or quadratic class label.

(a,b)	interval	label	(a,b)	interval	label
$(2,2)$	$(.043513, .043514)$	$[2,1]$	$(1,4)$	$(.079508, .079509)$	$P_+ : [1,1]$
$(1,3)$	$(.080124, .080125)$	$[3,0]$	$(1,2)$	$(.104688, .104689)$	$P_- : [1]$
$(1,1)$	$(.146404, .146405)$	$[4,1]$	$(0,6)$	$(.169377, .169378)$	$[3,1]$
$(0,5)$	$(.315227, .315228)$	$[1,1]$	$(0,4)$	$(.671533, .671534)$	$[2,0]$
$(0,3)$	$(1.489129, 1.489130)$	$[5,1]$	$(0,2)$	$(3.172313, 3.172314)$	$[4,0]$
$(0,1)$	$(5.903986, 5.903987)$	$[0,0]$	$(2,6)$	$(6.830385, 6.830386)$	$[1,0]$
$(3,6)$	$(9.552165, 9.552166)$	$P_- : [0]$	$(4,4)$	$(12.480646, 12.480647)$	$[0,1]$
$(3,5)$	$(12.577346, 12.577347)$	$P_+ : [0,0]$	$(4,5)$	$(22.981630, 22.981631)$	$[5,0]$

The split primes above 17 and 41 have ray logs $[1,0]$ and $[0,1]$ and act by automorphisms 5 and 10, respectively. Local Frobenius congruences certify these actions and every table label in `scripts/dimension_seven_artin_labels.gp`.

Rigorous Arb evaluation of the eight independent analytic values gives the raw logarithmic enclosure

$$\epsilon_{\log} \leq 5.86 \cdot 10^{-11}, \qquad 16128\epsilon_{\log} \leq 9.45 \cdot 10^{-7}.$$

The complete balls are archived in

`certificates/dimension-seven-double-sine-intervals.txt`.

Voutier's lower bound [5] is greater than $5.22 \cdot 10^{-5}$ for every relevant degree. The quotient of an analytic value by its assigned algebraic root must therefore be a root of unity; positivity and the exact cocycle multipliers select the displayed root.

2.5 Exact finite certificate

Let H be the degree-24 signed overlap field and $C = \mathbb{Q}(\zeta_{56})$. Their intersection has degree 12. The standard component is selected by the exact identity

$$c_H = \zeta_{56} + \zeta_{56}^{-1}, \qquad 1.98 < c_H < 1.99, \tag{16}$$

and it also satisfies

$$\sqrt{2} = \zeta_{56}^{7} + \zeta_{56}^{-7}. \tag{17}$$

In the resulting degree-48 compositum, `scripts/dimension_seven_exact_tcc.gp` verifies for each formal shift

$$\operatorname{Tr}\Pi = 1, \qquad \Pi^2 = \Pi, \qquad 441 \text{ rank-two minors vanish.} \tag{18}$$

Proposition 1 proves both shifts for the discriminant-32 tuple $Q_{7,2}$. The order of discriminant 32 has wide class number one. Unconditional $\mathrm{GL}_2(\mathbb{Z})$ covariance therefore transports the result to every admissible dimension-seven form of discriminant 32. Together with the independent discriminant-8 proof above, this proves the dimension-seven part of Theorem 2.

3 Dimension eight

Dimension-eight admissibility has $j = 2$ and $f_2 = 3$. The form conductor is therefore one or three, giving discriminants 5 and 45. These strata must be proved independently.

3.1 Conductor three: linear CM reinduction

For the canonical discriminant-45 form, the order-ray group is identified with the maximal-order ray group

$$\mathrm{Cl}_{(8)\infty_2}(\mathbb{Z}+3\mathcal{O}_{\mathbb{Q}(\sqrt{5})}) \simeq \mathrm{Cl}_{(24)\infty_2}(\mathcal{O}_{\mathbb{Q}(\sqrt{5})}) \simeq C_4 \times C_2^2. \tag{19}$$

Kopp's conductor-lowering formula supplies the exact three-factor dictionary. The fifteen lower characteristics lie in an index-two ray-12 field; Shintani exponent 576, Arb isolation, and height rigidity prove that entire lower packet.

The primitive packet contains two conjugate pairs of quartic characters. On the full two-place ray group, base conjugation does not literally send either one-place character to its inverse because it swaps the labeled infinite place. The projective quotient is nevertheless the same order-two character in both cases, and the associated biquadratic field has quadratic subfields

$$\mathbb{Q}(\sqrt{5}), \qquad \mathbb{Q}(\sqrt{-6}), \qquad \mathbb{Q}(\sqrt{-30}). \tag{20}$$

This projective observation is not used as a substitute for linear equivalence. The script `scripts/dimension_eight_linear_cm_reinduction.gp` compares the complete induced characters on the degree-16 normal closure and proves genuine linear reinduction, without a scalar twist, from quartic ray characters over $\mathbb{Q}(\sqrt{-6})$. The alternative base $\mathbb{Q}(\sqrt{-30})$ gives an independent check.

Lemma 3 (Ray labels and local factors)**.** *Let*

$$\mathfrak{m}_M = \mathfrak{p}_2^3\mathfrak{p}_3\mathfrak{p}_5,$$

with no infinite component, where the prime ideals are the exact factors printed by `scripts/dimension_eight_cm_unit_lattice.gp`*. The ray group* $\mathrm{Cl}_{\mathfrak{m}_M}(M)$ *has cyclic coordinates* $[8,4]$*, and in those coordinates the reinduced characters are*

$$\psi_0 = [6,1], \qquad \psi_1 = [2,1]. \tag{21}$$

They have the same complete Artin L-functions as the original $\mathbb{Q}(\sqrt{5})$ *characters* $[1,0,0]$ *and* $[1,1,0]$*, respectively. With*

$$S_M = \{\infty, \mathfrak{p}_2, \mathfrak{p}_3, \mathfrak{p}_5\}, \tag{22}$$

every finite member is ramified for ψ_b*, so* $L_{M,S_M}(s,\psi_b) = L_M(s,\psi_b)$*.*

Proof. The exact normal-closure calculation leaves precisely two faithful ray characters for each kernel, an inverse pair. For packet zero the coefficient at $n=5$ is $-i$ for $[6,1]$ and $+i$ for $[2,3]$; the source coefficient is $-i$. For packet one it is $+i$ for $[2,1]$ and $-i$ for $[6,3]$; the source coefficient is $+i$. This single exact separating coefficient selects among an exhaustive two-element set; it is not finite-coefficient recognition. `scripts/dimension_eight_cm_unit_lattice.gp` fixes the relative factors before constructing their ray groups and prints these four coefficients.

The two source characters have full conductor $(24)\infty_2$. The same audit proves that both CM conductors equal $\mathfrak{m}_M$ and that ψ_b is ramified at every finite member of S_M. Thus those local factors are already one. Linear reinduction gives the equality of the complete functions $L_M(s,\psi_b) = L_K(s,\chi_b)$; no unramified source Euler factor at 5 is removed. The archimedean parity is part of the same induced character equality. □

Lemma 4 (Normalized imaginary-quadratic Stark resolvent)**.** *Let* $M = \mathbb{Q}(\sqrt{-6})$ *and let* E_b/M *be either cyclic quartic extension selected in Lemma 3. There is a global unit* $\varepsilon_b \in E_b^\times$ *such that, for* $g \in \mathrm{Gal}(E_b/M)$*,*

$$\zeta'_{S_M}(0,g) = -\log|g(\varepsilon_b)|_{\mathrm{ord}}, \qquad |z|_{\mathrm{ord}} := \sqrt{z\bar{z}}. \tag{23}$$

Consequently

$$L'_{S_M}(0,\psi_b) = -\sum_g \overline{\psi_b(g)} \log|g(\varepsilon_b)|_{\mathrm{ord}}. \tag{24}$$

Proof. Stark's proved rank-one theorem for abelian extensions of an imaginary quadratic base [6] applies to E_b/M with the set (22). Here $|S_M| = 4 \geq 3$, so the unit clause in Stark's theorem gives a global unit rather than only an S-unit. The script `scripts/dimension_eight_cm_unit_lattice.gp` independently certifies `bnfcertify=1` and $e = |\mu(E_b)| = 2$ for both fields. Stark's normalized complex-place absolute value is $|z|^2_{\mathrm{ord}}$; hence its factor $1/e$ becomes coefficient one in the ordinary modulus convention used above. The finite set contains the distinct conductor primes above $2, 3, 5$. Fourier transforming the displayed partial-zeta formula gives (24). □

Exact unit-action matrices show that the quartic anti-unit component in Lemma 4 is a rank-two integral lattice. Rigorous Arb inversion isolates the four coordinate pairs

$$(0, 2), \quad (-2, 0), \quad (0, -2), \quad (2, 0) \tag{25}$$

with radii below $5 \cdot 10^{-9}$. Exact identities in the common normal closure then match their absolute norms to the original real-quadratic units. Thus the two oriented identities required by the AFK table hold exactly, rather than only in absolute value.

The signed packet and $\mathbb{Q}(\zeta_{16})$ lie in a convention-selected degree-128 compositum. `scripts/dimension_eight_exact_tcc.gp` verifies trace one, idempotency, and all 784 minors for both shifts.

3.2 Conductor one: quadratic units and exact phases

For $Q = \langle 1, -3, 1\rangle$, put

$$C = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_8 = C^6 = \begin{pmatrix} 377 & -144 \\ 144 & -55 \end{pmatrix}. \tag{26}$$

The one-place and both-place ray groups are C_2^2 and C_2^3. Kopp's exponent is one, and the difference is supported on two quadratic characters. Writing $\phi = (1 + \sqrt{5})/2$, their fields and oriented units are obtained from the positive roots

$$h = \sqrt{\phi} > 0, \qquad r = \sqrt{2\phi} > \phi :$$

	absolute equation	unit
L_0	$h^4 - h^2 - 1 = 0$	$u_0 = \phi + h$
L_1	$r^4 - 2r^2 - 4 = 0$	$u_1 = (r + \phi)/(r - \phi)$.

(27)

Both fields have class number one, with successful unconditional PARI certificates.

After dropping torsion, their fundamental-unit coordinates are

$$\phi = (-2, 0), \quad u_0 = (0, -1) \quad \text{in } L_0, \qquad \phi = (1, 0), \quad u_1 = (-1, -2) \quad \text{in } L_1.$$

Both determinant indices have absolute value two. This is the exact relative regulator-index calculation; the analytic class-number formula therefore gives

$$L'(0, \chi_0) = \log u_0, \qquad L'(0, \chi_1) = \log u_1. \tag{28}$$

Lemma 5 (Exact magnitude transform). *Let $c = (c_1, c_2)$ be the ray-8 class log of a primitive characteristic. Its differenced partial-zeta derivative is*

$$Z_c'(0) = \frac{1}{2}\left((-1)^{c_2} \log u_0 + (-1)^{c_1+c_2} \log u_1\right). \tag{29}$$

Hence, for

$$x^2 = u_0, \qquad d^2 = \sqrt{u_1/u_0}, \qquad a = xd, \tag{30}$$

the four ray-8 magnitudes are

$$[0,0] \mapsto a, \quad [0,1] \mapsto a^{-1}, \quad [1,0] \mapsto d^{-1}, \quad [1,1] \mapsto d. \tag{31}$$

At modulus four the stabilizer power is two and

$$Z_c'(0) = 2(-1)^c \log u_0, \tag{32}$$

giving magnitudes x^2, x^{-2}; at modulus two the magnitude is one. These cases exhaust all 63 *nonzero characteristics.*

Proof. For the ray group $G = C_2^2$, Fourier inversion gives

$$Z_c'(0) = \frac{1}{|G|} \sum_{\chi \in \widehat{G}} \overline{\chi(c)}(1 - \overline{\chi(R)}) L'(0, \chi).$$

The sign class is $R = [0,1]$, so only $\chi_0 = [0,1]$ and $\chi_1 = [1,1]$ survive, each with coefficient $1/2$. Substitution of (28) proves (29) and taking half the exponential gives (31). The identical calculation for the ray-4 group C_2, followed by the exactly audited stabilizer power two, proves (32). The characteristic audit prints 12 occurrences of each ray-8 label, six of each ray-4 label, and three modulus-two labels. □

The continued-fraction word of A_8 is $[3,3,3,3,3,3,0]$. Put $\beta = (3+\sqrt{5})/2$. The exact Euclidean recursion gives the rows

$$\begin{pmatrix} 377 & -144 \\ 144 & -55 \\ 55 & -21 \\ 21 & -8 \\ 8 & -3 \\ 3 & -1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \tag{33}$$

whose values on $(\beta, 1)^T$ are

$$\beta^7, \beta^6, \ldots, \beta, 1. \tag{34}$$

Thus all six adjacent period ratios in AFK's continued-fraction cocycle are β. For the characteristic $(p,q)/8$, division of the first period expression by the six successive denominators gives

$$z_i = \frac{q\beta^{i+2} - p\beta^{i+1}}{8}, \qquad 0 \le i < 6, \tag{35}$$

and the outer finite-product index is

$$\frac{-144p + (377-1)q}{8} = -18p + 47q. \tag{36}$$

These identities are checked in exact arithmetic by the archived maximal-tuple audit.

Put $e(t) = e^{2\pi i t}$ and

$$[z;\omega]_n = \begin{cases} \prod_{k=0}^{n-1}(1 - e(z + k\omega)), & n \ge 0, \\ \prod_{k=1}^{-n}(1 - e(z - k\omega))^{-1}, & n < 0. \end{cases} \tag{37}$$

For $s(z) = \lfloor -z \rfloor + 1$, define

$$\Sigma(z) = [z/\beta; -1/\beta]_{-s(z)} e\left(\frac{6(z+s(z))^2 + 6(1-\beta)(z+s(z))}{24\beta}\right) S_2(z + s(z) + 1 \mid \beta, 1). \tag{38}$$

Iterating AFK Definition 1.32 along the rows (33), inserting its outer finite product (36), and replacing each of Kopp's double sines by the reciprocal convention (1) gives

$$\nu_{p,q} = \frac{\tau^{-3(p^2-3pq+q^2)}(-1)^{(1+p)(1+q)}}{[(q\beta-p)/8;\beta]_{-18p+47q}} \prod_{i=0}^{5} \Sigma\left(\frac{q\beta^{i+2}-p\beta^{i+1}}{8}\right), \quad \tau = -e^{\pi i/8}. \tag{39}$$

For real $t \notin \mathbb{Z}$,

$$1 - e^{2\pi i t} = -2ie^{\pi i t}\sin(\pi t). \tag{40}$$

The fundamental reciprocal double sine is positive on $0 < z < \beta+1$, since it is a ratio of positive Barnes double-gamma values. Equations (2), (39), and (40) therefore reduce every phase to an exact element of $\mathbb{Q}(\beta)/2\mathbb{Z}$. `scripts/dimension_eight_maximal_sign_audit.py` proves for all 63 nonzero characteristics that the coefficient of β cancels, the remaining coefficient of π is integral, and its parity agrees with the signed radical table. Decimal arithmetic is used only to guess a floor; both defining inequalities are then verified exactly in $\mathbb{Q}(\sqrt{5})$. The audit also proves that no finite-product factor or recurrence sine vanishes.

With x, d, a as in Lemma 5, the table uses only

$$\pm 1, \quad \pm x^2, \quad \pm x^{-2}, \quad \pm d, \quad \pm d^{-1}, \quad \pm a, \quad \pm a^{-1}. \tag{41}$$

Exact shared-subfield identities glue it to $\mathbb{Q}(\zeta_{16})$. `scripts/dimension_eight_maximal_exact_tcc.py` verifies, for each shift,

$$\operatorname{Tr}\Pi = 1, \qquad \Pi^2 = \Pi, \qquad \text{784 rank-two minors vanish.} \tag{42}$$

3.3 Form-class completion

Solving the admissibility recurrences gives

$$d = 8 \Longrightarrow r = 1, \quad n = 9, \quad K = \mathbb{Q}(\sqrt{5}), \quad j = 2, \quad m = 1, \quad f_2 = 3. \tag{43}$$

Hence the form conductor is exactly $f = 1$ or $f = 3$, and the only discriminants are 5 and 45. The exact wide class calculations are

$$h_{\mathrm{GL}_2}(5) = h_{\mathrm{GL}_2}(45) = 1. \tag{44}$$

For discriminant five, the maximal order also has narrow class number one because ϕ has norm -1. For discriminant 45 only the wide ring class number is used: determinant--1 transformations are allowed, and no narrow-class-number assertion is needed.

If t_M is obtained from t by $M \in \mathrm{GL}_2(\mathbb{Z})$, the unconditional covariance calculation of the companion Paper I [3] sends the complete TCC sum at $(t_M, \boldsymbol{p})$ to that at $(t, \ell M\boldsymbol{p})$, after reindexing its representatives; for $\det M = -1$ both cocycle factors are conjugated. This proves $\mathcal{Z}_t \subseteq \mathcal{Z}_{t_M}$. Applying the same calculation to M^{-1} proves the reverse inclusion. Equations (43)–(44) therefore transport each of the two certified representatives across its own discriminant and complete dimension eight.

Proof of Theorem 2. Section 2 proves both shifts independently for the discriminant-8 and discriminant-32 dimension-seven representatives, then transports within each discriminant. Section 3 proves both dimension-eight conductors and transports within each of the only two possible discriminants. □

4 Why dimension six remains different

The dimension-six primitive character has order six. Its projective quotient has order three, rather than the order-two quotient that produces the V_4 CM descent in dimension eight. The alternative index-two induction groups are nonabelian, so no second abelian induction base supplies an imaginary-quadratic elliptic-unit theorem. The finite dimension-six candidate already satisfies all 225 minors; the remaining input is one oriented primitive order-six special value. Thus the gap is analytic, not a finite reconstruction problem.

5 Reproducibility

The final audits were run on Linux 6.8.0-124-generic (x86-64, four virtual AMD EPYC 9354P cores) with PARI/GP 2.15.4, Python 3.12.3, NumPy 1.26.4, python-flint 0.9.0, and FLINT 3.6.0. Representative measurements on that virtual machine were

audit	wall time	peak RSS
dimension-seven closure suite	43.5 s	179 MB
Paper-II archive smoke suite	58.6 s	179 MB.

The submission-specific archive is built deterministically, carries a self-verifying root manifest named `ARCHIVE_CONTENTS.sha256`, and passes its tests after clean extraction. Its Zenodo DOI is `https://doi.org/10.5281/zenodo.21681700`.

The supplementary archive contains:

- all dimension-seven maximal-order and conductor-lowering, Shintani-divisor, Arb-height, exact-phase, Artin-label, field-gluing, and exact-minor scripts, including the complete divisor and enclosure transcripts;
- all conductor-three dimension-eight order/maximal-order, linear-reinduction, CM-unit, Arb-orientation, normal-closure, and exact-minor scripts;
- all maximal-order dimension-eight ray, quadratic unit, exact phase, field gluing, and exact minor scripts;
- generated CM-orientation and 63-sign transcripts, the global regression transcript, and SHA-256 checksums; and
- end-to-end regression tests for both dimension-eight strata.

The principal reproduction commands are

```
python3 -m unittest tests.test_dimension_seven_closure
gp -q scripts/dimension_seven_admissible_strata.gp
gp -q scripts/dimension_seven_exact_tcc.gp
python3 scripts/certify_dimension_seven_double_sine.py --tolerance 1e-10
gp -q scripts/dimension_eight_linear_cm_reinduction.gp
gp -q scripts/dimension_eight_cm_unit_lattice.gp
python3 scripts/certify_dimension_eight_cm_orientation.py
gp -q scripts/dimension_eight_cm_real_unit_bridge.gp
gp -q scripts/dimension_eight_exact_tcc.gp
gp -q scripts/dimension_eight_maximal_tuple_audit.gp
gp -q scripts/dimension_eight_maximal_quadratic_units.gp
python3 scripts/dimension_eight_maximal_sign_audit.py
python3 scripts/dimension_eight_maximal_exact_tcc.py
```

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